

Signal and Systems Analysis Exam Solutions

Michelle L. Wu

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Signal and Systems Analysis
Taught by: Dr. Dr. Vasilis Marmarelis

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1 Exam 1

1.1 Problem 1

Find the expansion of the function:

$$x(t) = (2 + 3t^3)$$

on the basis $\{1, t\}$ over $t \in [-1, 1]$, so that the approximation mean-square error be minimum.

1.1.1 Solution

Proof. Let $x(t) = 2 + 3t^3 \in L^2([-1, 1])$. Consider the subspace $V = \text{span}\{1, t\} \subset L^2([-1, 1])$, which is a closed subspace of a Hilbert space.

By the *Orthogonal Projection Theorem* (see e.g., Kreyszig, *Introductory Functional Analysis with Applications*, Theorem 6.2-1), there exists a unique element $\hat{x}(t) \in V$ that minimizes the mean-square error

$$\|x - \hat{x}\|^2 = \int_{-1}^1 [x(t) - \hat{x}(t)]^2 dt,$$

and satisfies the orthogonality conditions:

$$\langle x - \hat{x}, v \rangle = 0 \quad \forall v \in V.$$

Denote $\hat{x}(t) = a_0 + a_1 t$. Then, the orthogonality conditions yield the system:

$$\langle x - (a_0 + a_1 t), 1 \rangle = 0, \quad \langle x - (a_0 + a_1 t), t \rangle = 0.$$

Define the inner product on $L^2([-1, 1])$ as:

$$\langle f, g \rangle := \int_{-1}^1 f(t)g(t) dt.$$

Compute as follows.

$$\begin{aligned} \langle x, 1 \rangle &= \int_{-1}^1 (2 + 3t^3) dt = 2 \cdot 2 + 0 = 4, \\ \langle x, t \rangle &= \int_{-1}^1 (2t + 3t^4) dt = 0 + 3 \cdot \int_{-1}^1 t^4 dt = 3 \cdot \frac{2}{5} = \frac{6}{5}, \\ \langle 1, 1 \rangle &= \int_{-1}^1 1 \cdot 1 dt = 2, \\ \langle t, t \rangle &= \int_{-1}^1 t^2 dt = \frac{2}{3} \end{aligned}$$

Using the projection formula:

$$a_0 = \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} = \frac{4}{2} = 2, \quad a_1 = \frac{\langle x, t \rangle}{\langle t, t \rangle} = \frac{\frac{6}{5}}{\frac{2}{3}} = \frac{18}{10} = \frac{9}{5}.$$

Thus, the best least-squares approximation to $x(t)$ in the subspace V is:

$$\hat{x}(t) = 2 + \frac{9}{5}t,$$

and this function minimizes the L^2 -norm $\|x - \hat{x}\|$ over all functions in V . □

1.2 Problem 2

Find the Fourier Series expansion of the function $f(t)$ over $t \in [-1, 1]$ given by:

$$f(t) = |\cos(\pi t)| + 1.$$

Find the Fourier Transform of the periodic function with period $f(t)$ from minus to plus infinity.

1.2.1 Solution

Proof. Let $f(t) = |\cos(\pi t)| + 1$, which is an even, periodic function with fundamental period $T = 2$. Hence, the fundamental angular frequency is $\omega_0 = \pi$. Since $f(t)$ is even, its Fourier series contains only cosine terms.

Denote the real Fourier series representation as

$$f(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t),$$

where the coefficients are given by

$$a_k = \frac{2}{T} \int_0^{T/2} f(t) \cos(k\omega_0 t) dt = \int_0^1 f(t) \cos(k\pi t) dt.$$

Since $f(t) = 1 + |\cos(\pi t)|$, we can decompose:

$$a_k = \int_0^1 \cos(k\pi t) dt + \int_0^1 |\cos(\pi t)| \cos(k\pi t) dt.$$

The average (DC) term is

$$a_0 = \frac{1}{T} \int_{-1}^1 f(t) dt = \frac{1}{2} \left(\int_{-1}^1 |\cos(\pi t)| dt + \int_{-1}^1 1 dt \right) = \frac{1}{2} \left(\frac{4}{\pi} + 2 \right) = 1 + \frac{2}{\pi}.$$

The function $|\cos(\pi t)|$ uses the known cosine series expansion:

$$|\cos(\pi t)| = \frac{2}{\pi} + \sum_{k=1}^{\infty} \frac{4(-1)^k}{\pi(1-4k^2)} \cos(2k\pi t).$$

Multiplying by 1 and adding the DC offset, we obtain

$$f(t) = 1 + \frac{2}{\pi} + \sum_{k=1}^{\infty} \frac{4(-1)^k}{\pi(1-4k^2)} \cos(2k\pi t).$$

Equivalently, in the exponential form $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$ with $\omega_0 = \pi$ and $T = 2$, the coefficients satisfy

$$c_0 = 1 + \frac{2}{\pi}, \quad c_{\pm 2k} = \frac{2(-1)^k}{\pi(1 - 4k^2)}, \quad c_n = 0 \text{ for all odd } n.$$

For a periodic signal $f(t)$ with period T and coefficients c_n , its Fourier transform (in the distributional sense) is given by

$$\mathcal{F}\{f\}(\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0).$$

Substituting $\omega_0 = \pi$ and the computed c_n , we obtain

$$\mathcal{F}\{f\}(\omega) = (2\pi + 4) \delta(\omega) + \sum_{k=1}^{\infty} \frac{4(-1)^k}{1 - 4k^2} [\delta(\omega - 2k\pi) + \delta(\omega + 2k\pi)].$$

□

1.3 Problem 3

Using the Fourier Transform, find the function $y(t)$ that satisfies the differential equation:

$$\frac{d^2y}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = \delta(t) + e^{-t}u(t).$$

Then, find the Fourier Transform of the function $x(t)$ defined as:

$$x(t) = \frac{dy(t)}{dt} + 2y(t)$$

1.3.1 Solution

Proof. Consider the linear constant-coefficient differential equation

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y(t) = \delta(t) + e^{-t}u(t),$$

with zero initial conditions. Let $Y(\omega) = \mathcal{F}\{y(t)\}$.

Using standard Fourier transform identities for derivatives, we have

$$\begin{aligned}\mathcal{F}\left\{\frac{d^2y}{dt^2}\right\}(\omega) &= -(i\omega)^2Y(\omega) = -\omega^2Y(\omega), \\ \mathcal{F}\left\{\frac{dy}{dt}\right\}(\omega) &= i\omega Y(\omega), \\ \mathcal{F}\{y(t)\}(\omega) &= Y(\omega).\end{aligned}$$

Applying the Fourier transform to both sides of the ODE gives

$$(-\omega^2 + 6i\omega + 8)Y(\omega) = \mathcal{F}\{\delta(t)\} + \mathcal{F}\{e^{-t}u(t)\}.$$

Using standard results in the sense of tempered distributions:

$$\mathcal{F}\{\delta(t)\} = 1, \quad \mathcal{F}\{e^{-t}u(t)\} = \frac{1}{1+i\omega}.$$

Thus,

$$(-\omega^2 + 6i\omega + 8)Y(\omega) = 1 + \frac{1}{1+i\omega}.$$

Solving for $Y(\omega)$ results in:

$$Y(\omega) = \frac{1 + \frac{1}{1+i\omega}}{-\omega^2 + 6i\omega + 8} = \frac{(2+i\omega)}{(1+i\omega)(-\omega^2 + 6i\omega + 8)}.$$

Note that

$$-\omega^2 + 6i\omega + 8 = -(\omega^2 - 6i\omega - 8) = -(\omega - 2i)(\omega - 4i).$$

Simplifying the denominator results in:

$$Y(\omega) = -\frac{(2 + i\omega)}{(1 + i\omega)(\omega - 2i)(\omega - 4i)}.$$

By linearity of the Fourier transform,

$$\mathcal{F}\{x(t)\} = \mathcal{F}\{y'(t) + 2y(t)\} = (i\omega + 2)Y(\omega).$$

Therefore,

$$\mathcal{F}\{x(t)\} = -\frac{(i\omega + 2)(2 + i\omega)}{(1 + i\omega)(\omega - 2i)(\omega - 4i)}.$$

□

1.4 Problem 4

Using the Laplace Transform, find the solution of the following system of differential equations:

$$\begin{aligned}\frac{dy(t)}{dt} + 2y(t) &= x(t) + \delta(t) \\ \frac{dx(t)}{dt} + x(t) &= 2y(t)\end{aligned}$$

for zero initial conditions, where $u(t)$ denotes the step function.

1.4.1 Solution

Proof. Let:

$$X(s) = \mathcal{L}\{x(t)\}, \quad Y(s) = \mathcal{L}\{y(t)\}.$$

Apply $\mathcal{L}$ to both equations.

Define the following as Equation (1).

$$\mathcal{L}\left\{\frac{dy}{dt} + 2y(t)\right\} = sY(s) + 2Y(s) = X(s) + \mathcal{L}\{\delta(t)\} = X(s) + 1.$$

So,

$$(s + 2)Y(s) = X(s) + 1 \tag{1}$$

Now, define the following as Equation (2).

$$\mathcal{L}\left\{\frac{dx}{dt} + x(t)\right\} = sX(s) + X(s) = (s + 1)X(s) = 2Y(s). \tag{2}$$

From (2), express $X(s)$ in terms of $Y(s)$:

$$X(s) = \frac{2}{s + 1}Y(s). \tag{3}$$

Substitute (3) into (1):

$$(s + 2)Y(s) = \frac{2}{s + 1}Y(s) + 1.$$

Multiply through by $(s + 1)$ to eliminate the denominator:

$$(s + 2)(s + 1)Y(s) = 2Y(s) + (s + 1).$$

Now expanding:

$$(s^2 + 3s + 2)Y(s) - 2Y(s) = s + 1.$$

So,

$$(s^2 + 3s)Y(s) = s + 1 \Rightarrow Y(s) = \frac{s + 1}{s(s + 3)}. \quad (4)$$

From (3),

$$X(s) = \frac{2}{s + 1} \cdot \frac{s + 1}{s(s + 3)} = \frac{2}{s(s + 3)}. \quad (5)$$

Now, invert both (4) and (5).

For $Y(s) = \frac{s+1}{s(s+3)}$, by partial fraction decomposition:

$$\frac{s + 1}{s(s + 3)} = \frac{A}{s} + \frac{B}{s + 3}.$$

Multiply both sides by $s(s + 3)$:

$$s + 1 = A(s + 3) + Bs.$$

Expanding as follows.

$$s + 1 = As + 3A + Bs = (A + B)s + 3A.$$

Matching coefficients as follows.

$$\begin{cases} A + B = 1, \\ 3A = 1 \Rightarrow A = \frac{1}{3}, \quad B = \frac{2}{3} \end{cases}$$

Thus,

$$Y(s) = \frac{1}{3s} + \frac{2}{3(s + 3)}.$$

Take inverse transforms as follows.

$$y(t) = \mathcal{L}^{-1} \left[\frac{1}{3s} \right] + \mathcal{L}^{-1} \left[\frac{2}{3(s + 3)} \right] = \frac{1}{3}u(t) + \frac{2}{3}e^{-3t}u(t)$$

For $X(s) = \frac{2}{s(s+3)}$, use

$$\frac{2}{s(s + 3)} = \frac{C}{s} + \frac{D}{s + 3}.$$

Multiply as follows.

$$2 = C(s + 3) + Ds \Rightarrow 2 = (C + D)s + 3C$$

Matching coefficients as follows.

$$\begin{cases} C + D = 0 \Rightarrow D = -C, \\ 3C = 2 \Rightarrow C = \frac{2}{3}, \quad D = -\frac{2}{3} \end{cases}$$

Then,

$$X(s) = \frac{2}{3s} - \frac{2}{3(s+3)} \quad \Rightarrow \quad x(t) = \frac{2}{3}u(t) - \frac{2}{3}e^{-3t}u(t).$$

Therefore,

$$\begin{aligned} y(t) &= \frac{1}{3}u(t) + \frac{2}{3}e^{-3t}u(t), \\ x(t) &= \frac{2}{3}u(t) - \frac{2}{3}e^{-3t}u(t), \end{aligned} \quad t \geq 0.$$

□

1.5 Problem 5

Find the discrete-time causal solution $y(n)$ of the Difference Equation:

$$y(n) - \frac{11}{15}y(n-1) + \frac{2}{15}y(n-2) = x(n),$$

where $x(n) = (\frac{1}{2})^n u(n)$ and $u(n)$ denotes the discrete time step function.

1.5.1 Solution

Proof. Given the second-order linear constant-coefficient difference equation:

$$y(n) - \frac{11}{15}y(n-1) + \frac{2}{15}y(n-2) = x(n), \quad x(n) = \left(\frac{1}{2}\right)^n u(n).$$

Let $Y(z) = \mathcal{Z}\{y(n)\}$ and $X(z) = \mathcal{Z}\{x(n)\}$. Applying the Z -transform, assuming zero initial conditions, and using the time-shift property results in:

$$Y(z) \left(1 - \frac{11}{15}z^{-1} + \frac{2}{15}z^{-2}\right) = X(z).$$

Since $x(n) = (\frac{1}{2})^n u(n)$, the following is obtained.

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

Hence,

$$Y(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - \frac{11}{15}z^{-1} + \frac{2}{15}z^{-2})}.$$

Define $w = z^{-1}$. Then:

$$Y(z) = \frac{1}{(1 - \frac{1}{2}w)(1 - \frac{11}{15}w + \frac{2}{15}w^2)}.$$

The quadratic factor can be rewritten as:

$$1 - \frac{11}{15}w + \frac{2}{15}w^2 = (1 - r_1w)(1 - r_2w),$$

where r_1, r_2 are the roots of

$$r^2 - \frac{11}{15}r + \frac{2}{15} = 0.$$

Solving for r yields:

$$r_{1,2} = \frac{\frac{11}{15} \pm \sqrt{\left(\frac{11}{15}\right)^2 - 4\left(\frac{2}{15}\right)}}{2} = \frac{11 \pm 1}{30} = \left\{\frac{2}{5}, \frac{1}{3}\right\}.$$

Hence,

$$Y(z) = \frac{1}{(1 - \frac{1}{2}w)(1 - \frac{2}{5}w)(1 - \frac{1}{3}w)}.$$

By partial fraction decomposition, denote $Y(z)$ as:

$$Y(z) = \frac{A}{1 - \frac{1}{2}w} + \frac{B}{1 - \frac{2}{5}w} + \frac{C}{1 - \frac{1}{3}w}.$$

Using the cover-up method at the simple poles $w = 2, \frac{5}{2}, 3$ (reciprocal of each factor):

$$\begin{aligned} A &= \frac{1}{(1 - \frac{2}{5} \cdot \frac{1}{2})(1 - \frac{1}{3} \cdot \frac{1}{2})} = 15, \\ B &= \frac{1}{(1 - \frac{1}{2} \cdot \frac{5}{2})(1 - \frac{1}{3} \cdot \frac{5}{2})} = 10, \\ C &= \frac{1}{(1 - \frac{1}{2} \cdot 3)(1 - \frac{2}{5} \cdot 3)} = -24. \end{aligned}$$

Thus,

$$Y(z) = 15 \cdot \frac{1}{1 - \frac{1}{2}z^{-1}} + 10 \cdot \frac{1}{1 - \frac{1}{3}z^{-1}} - 24 \cdot \frac{1}{1 - \frac{2}{5}z^{-1}}.$$

Since

$$\mathcal{Z}^{-1} \left\{ \frac{1}{1 - az^{-1}} \right\} = a^n u(n),$$

the following is obtained.

$$y(n) = 15 \left(\frac{1}{2} \right)^n u(n) + 10 \left(\frac{1}{3} \right)^n u(n) - 24 \left(\frac{2}{5} \right)^n u(n)$$

Therefore,

$$y(n) = \left[15 \left(\frac{1}{2} \right)^n + 10 \left(\frac{1}{3} \right)^n - 24 \left(\frac{2}{5} \right)^n \right] u(n).$$

□

1.5.2 Bonus Problem

This section is included since I made a mistake in rewriting the problem and wondered why the numbers were so intriguing.

Find the discrete-time causal solution $y(n)$ of the Difference Equation:

$$y(n) - \frac{11}{5}y(n-1) + \frac{2}{15}y(n-2) = x(n),$$

where $x(n) = (\frac{1}{2})^n u(n)$ and $u(n)$ denotes the discrete time step function.

1.5.3 Solution

Proof. Given the second-order linear difference equation:

$$y(n) - \frac{11}{5}y(n-1) + \frac{2}{15}y(n-2) = x(n), \quad x(n) = \left(\frac{1}{2}\right)^n u(n).$$

Let $Y(z) = \mathcal{Z}\{y(n)\}$ and $X(z) = \mathcal{Z}\{x(n)\}$. Applying the Z-transform (zero initial conditions) and using time-shift:

$$Y(z) - \frac{11}{5}z^{-1}Y(z) + \frac{2}{15}z^{-2}Y(z) = X(z).$$

Factor out $Y(z)$:

$$Y(z) \left(1 - \frac{11}{5}z^{-1} + \frac{2}{15}z^{-2}\right) = X(z).$$

Since $x(n) = (\frac{1}{2})^n u(n)$, we have

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}.$$

Substitute into the previous equation:

$$Y(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{11}{5}z^{-1} + \frac{2}{15}z^{-2}\right)}.$$

Let $w := z^{-1}$. The quadratic factor in w allows the factorization

$$1 - \frac{11}{5}w + \frac{2}{15}w^2 = (1 - r_1 w)(1 - r_2 w),$$

where $r_{1,2}$ are the roots of $r^2 - \frac{11}{5}r + \frac{2}{15} = 0$, i.e.

$$r_{1,2} = \frac{33 \pm \sqrt{969}}{30} \quad (r_1 \approx 2.1376, r_2 \approx 0.06237).$$

Hence,

$$Y(z) = \frac{1}{\left(1 - \frac{1}{2}w\right) (1 - r_1 w) (1 - r_2 w)}.$$

By partial fraction decomposition, denote

$$Y(z) = \frac{A}{1 - \frac{1}{2}w} + \frac{B}{1 - r_1 w} + \frac{C}{1 - r_2 w}.$$

Using the cover-up method directly in w (evaluate at the simple poles $w = \frac{1}{1/2}, \frac{1}{r_1}, \frac{1}{r_2}$), we obtain

$$A = \frac{1}{\left(1 - \frac{r_1}{1/2}\right)\left(1 - \frac{r_2}{1/2}\right)} = \frac{1}{(1 - 2r_1)(1 - 2r_2)} = \frac{1}{1 - 2(r_1 + r_2) + 4r_1 r_2} = -\frac{15}{43},$$

$$B = \frac{1}{\left(1 - \frac{1}{2r_1}\right)\left(1 - \frac{r_2}{r_1}\right)} = \frac{2r_1^2}{(2r_1 - 1)(r_1 - r_2)}, \quad C = \frac{1}{\left(1 - \frac{1}{2r_2}\right)\left(1 - \frac{r_1}{r_2}\right)} = \frac{2r_2^2}{(2r_2 - 1)(r_2 - r_1)}.$$

Numerically,

$$A = -\frac{15}{43} \approx -0.34884, \quad B \approx 1.34455, \quad C \approx 0.004284.$$

Since $\mathcal{Z}^{-1}\left\{\frac{1}{1-az^{-1}}\right\} = a^n u(n)$, we obtain

$$y(n) = A \left(\frac{1}{2}\right)^n u(n) + B r_1^n u(n) + C r_2^n u(n).$$

Thus,

$$y(n) = \left(-\frac{15}{43} \left(\frac{1}{2}\right)^n + 1.34455 r_1^n + 0.004284 r_2^n\right) u(n), \quad r_1 \approx 2.1376, \quad r_2 \approx 0.06237.$$

To check convergence and stability, the poles are at $z = \frac{1}{2}, r_1, r_2$. For a causal sequence,

$$\text{ROC: } |z| > \max\left\{\frac{1}{2}, r_1, r_2\right\} = r_1 \approx 2.1376.$$

Since $r_1 > 1$, the response is unstable. □

2 Exam 2

2.1 Problem 1

You are given the system configuration shown below, where the subsystems A, B, and C are described by the following linear differential equations:

$$\text{A: } \dot{y} + y = x \quad \text{B: } \dot{y} + 4y = \dot{x} + x \quad \text{C: } \dot{y} + 3y = x$$

where x and y denote the putative input and output, respectively.

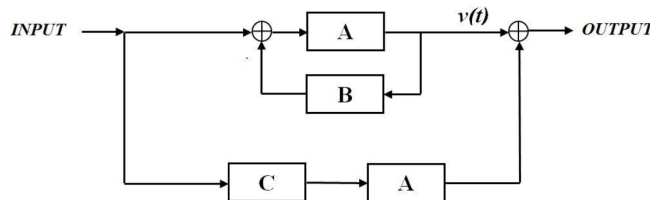


Figure 1

Find the following.

1. The Transfer Function and IRF of the overall system.
2. The system response to a step input.
3. The system response to a non-causal sinusoidal input: $\cos(\alpha t)$.
4. The amplitude PDF, the autocorrelation function and the spectrum of the system output, when the input is Gaussian White Noise with unity power level.
5. The input-output cross-correlation and cross-spectrum for the same input in Q4.
6. The amplitude PDF and spectrum of $v(t)$ (marked on diagram) for same input in Q4

2.1.1 Solution

With zero initial conditions, Laplace transforms yield

$$A(s) = \frac{1}{s+1}, \quad B(s) = \frac{s+1}{s+4}, \quad C(s) = \frac{1}{s+3}.$$

Let $v(t)$ be the signal marked in the diagram (output of the upper A), and let $y(t)$ be the overall output.

The upper path is a negative-feedback loop: the summer forms $e(t) = x(t) - b(t)$ where $b(t) = (B * v)(t)$, and then $v(t) = (A * e)(t)$. In the Laplace domain,

$$V(s) = A(s)(X(s) - B(s)V(s)).$$

Proof. Rearranging $V = A(X - BV)$ gives $V(1 + AB) = AX$, hence $V/X = A/(1 + AB)$. This is the standard closed-loop formula for negative feedback. $\square$

Therefore, the transfer from input to the internal node $v(t)$ is

$$G_v(s) := \frac{V(s)}{X(s)} = \frac{A(s)}{1 + A(s)B(s)} = \frac{\frac{1}{s+1}}{1 + \frac{1}{s+1} \frac{s+1}{s+4}} = \frac{\frac{1}{s+1}}{1 + \frac{1}{s+4}} = \frac{s+4}{(s+1)(s+5)}.$$

The lower branch is a cascade C then A , so it contributes $A(s)C(s) = \frac{1}{(s+1)(s+3)}$. The output summer adds the two branches, hence

$$H(s) := \frac{Y(s)}{X(s)} = G_v(s) + A(s)C(s) = \frac{s+4}{(s+1)(s+5)} + \frac{1}{(s+1)(s+3)}.$$

Equivalently,

$$H(s) = \frac{s^2 + 8s + 17}{(s+1)(s+3)(s+5)}.$$

To invert, write the partial fraction form

$$H(s) = \frac{5}{4(s+1)} - \frac{1}{2(s+3)} + \frac{1}{4(s+5)}.$$

Thus, the impulse response $h(t) = \mathcal{L}^{-1}\{H(s)\}$ is

$$h(t) = \frac{5}{4}e^{-t}u(t) - \frac{1}{2}e^{-3t}u(t) + \frac{1}{4}e^{-5t}u(t).$$

The step response is $y_{\text{step}}(t) = \mathcal{L}^{-1}\{H(s)/s\}$, giving

$$y_{\text{step}}(t) = \frac{17}{15} - \frac{5}{4}e^{-t} + \frac{1}{6}e^{-3t} - \frac{1}{20}e^{-5t}, \quad t \geq 0.$$

Indeed $y_{\text{step}}(\infty) = H(0) = \frac{17}{15}$. Moreover, $y_{\text{step}}(0^+) = \lim_{s \rightarrow \infty} H(s) = 0$, consistent with a strictly proper stable transfer function.

For a stable LTI system driven by the two-sided sinusoid $x(t) = \Re\{e^{j\alpha t}\}$, the sinusoidal steady-state (i.e., particular) response is

$$y(t) = \Re\{H(j\alpha)e^{j\alpha t}\} = |H(j\alpha)| \cos(\alpha t + \angle H(j\alpha)),$$

where

$$H(j\alpha) = \frac{(j\alpha)^2 + 8(j\alpha) + 17}{(j\alpha + 1)(j\alpha + 3)(j\alpha + 5)}.$$

Continued on the next page.

Proof. Write $\cos(\alpha t) = \Re\{e^{j\alpha t}\}$. By frequency-response theory for LTI systems, $e^{j\alpha t} \mapsto H(j\alpha)e^{j\alpha t}$. Taking the real part yields the stated magnitude/phase form. $\square$

Assume $x(t)$ is zero-mean Gaussian white noise with two-sided PSD $S_{xx}(\omega) = 1$ (equivalently $R_{xx}(\tau) = \delta(\tau)$). Here, white noise is understood in the standard continuous-time generalized stochastic distribution sense.

Adopt the two-sided Wiener–Khinchin/Fourier convention

$$R_{xx}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{j\omega\tau} d\omega, \quad S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau,$$

under which $S_{xx}(\omega) = 1$ corresponds to $R_{xx}(\tau) = \delta(\tau)$.

Since y is a linear transform of a Gaussian process, $y(t)$ is Gaussian:

$$y(t) \sim \mathcal{N}(0, \sigma_y^2).$$

Hence, the amplitude PDF is

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{y^2}{2\sigma_y^2}\right).$$

Write $H(s) = N(s)/D(s)$ with $N(s) = s^2 + 8s + 17$ and $D(s) = (s+1)(s+3)(s+5)$. Then

$$S_{yy}(\omega) = |H(j\omega)|^2 = \frac{N(j\omega)N(-j\omega)}{D(j\omega)D(-j\omega)} = \frac{\omega^4 + 30\omega^2 + 289}{(\omega^2 + 1)(\omega^2 + 9)(\omega^2 + 25)}.$$

The autocorrelation is the inverse Fourier transform

$$R_{yy}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{yy}(\omega) e^{j\omega\tau} d\omega,$$

and for a causal stable impulse response h ,

$$R_{yy}(\tau) = \int_0^{\infty} h(t) h(t + |\tau|) dt, \quad R_{yy}(-\tau) = R_{yy}(\tau).$$

In particular,

$$\sigma_y^2 = R_{yy}(0) = \int_0^{\infty} h^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{yy}(\omega) d\omega.$$

Define $R_{xy}(\tau) = \mathbb{E}[x(t)y(t+\tau)]$.

Note that this definition, R_{xy} is a forward correlation in y ; some texts use $\mathbb{E}[x(t+\tau)y(t)]$, which would shift signs accordingly.

Proof. Since $y = h * x$, we have $y(t + \tau) = \int_{-\infty}^{\infty} h(u)x(t + \tau - u) du$. Taking expectations and using linearity gives

$$R_{xy}(\tau) = \mathbb{E}[x(t)y(t + \tau)] = \int_{-\infty}^{\infty} h(u) \mathbb{E}[x(t)x(t + \tau - u)] du = \int_{-\infty}^{\infty} h(u)R_{xx}(\tau - u) du.$$

With $R_{xx}(\tau) = \delta(\tau)$, it follows that

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} h(u)\delta(\tau - u) du = h(\tau).$$

□

Here, $h(\tau)$ is understood as the two-sided impulse response, i.e., $h(\tau) = 0$ for $\tau < 0$ due to causality. The cross-spectrum is

$$S_{xy}(\omega) = H(j\omega)S_{xx}(\omega) = H(j\omega), \quad S_{yx}(\omega) = H^*(j\omega).$$

Under the same Fourier convention (and for real x, y), the cross-spectra satisfy $S_{xy}(\omega) = H(j\omega)S_{xx}(\omega)$ and $S_{yx}(\omega) = S_{xy}^*(\omega)$.

The internal node $v(t)$ has transfer

$$G_v(s) = \frac{V}{X} = \frac{s + 4}{(s + 1)(s + 5)}.$$

Thus,

$$S_{vv}(\omega) = |G_v(j\omega)|^2 S_{xx}(\omega) = \frac{\omega^2 + 16}{(\omega^2 + 1)(\omega^2 + 25)}.$$

Since v is also a linear transform of Gaussian white noise,

$$v(t) \sim \mathcal{N}(0, \sigma_v^2), \quad f_V(v) = \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{v^2}{2\sigma_v^2}\right),$$

with

$$\sigma_v^2 = R_{vv}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{vv}(\omega) d\omega.$$

2.2 Extra Credit

You are given the cascade of an LTI system with step response:

$$(1 - e^{-t}) \cdot u(t)$$

followed by a static nonlinearity:

$$y(t) = v(t) + v^2(t),$$

where $v(t)$ denotes the output of the LTI system. Find the following.

2.2.1 Problem 1

The response of this cascade system to an impulse input:

$$x(t) = A \cdot \delta(t).$$

2.2.1.1 Solution For a causal LTI system, the step response is the integral of the impulse response:

$$s(t) = \int_0^t h(\lambda) d\lambda.$$

Differentiating (in the distributional sense) gives

$$h(t) = \frac{d}{dt}s(t) = \frac{d}{dt}((1 - e^{-t})u(t)) = e^{-t}u(t).$$

Proof. Since $s(t) = (h * u)(t) = \int_0^t h(\lambda) d\lambda$ for causal h , differentiation yields $h(t) = \frac{d}{dt}s(t)$. Here, $s(t) = (1 - e^{-t})u(t)$ is continuous at $t = 0$ with $s(0) = 0$, so no additional impulse term appears at $t = 0$, giving $h(t) = e^{-t}u(t)$. $\square$

With $x(t) = A\delta(t)$,

$$v(t) = (h * x)(t) = Ah(t) = Ae^{-t}u(t).$$

Then,

$$y(t) = v(t) + v^2(t) = Ae^{-t}u(t) + A^2e^{-2t}u(t),$$

since $u^2(t) = u(t)$ almost everywhere. This equality holds almost everywhere (and in the distributional sense). Since the input $x(t) = A\delta(t)$ is deterministic and the nonlinearity is pointwise, static, the result follows without invoking any probabilistic assumptions.

2.2.2 Problem 2

The input-output cross-correlation of this cascade system when the input is Gaussian White Noise with unity power level.

2.2.2.1 Solution From Problem 1, the impulse response is $h(t) = e^{-t}u(t)$, and

$$v(t) = (h * x)(t) = \int_{-\infty}^{\infty} h(\lambda) x(t - \lambda) d\lambda.$$

The cascade output is $y(t) = v(t) + v^2(t)$.

Adopt the convention

$$R_{xy}(\tau) = \mathbb{E}[x(t)y(t + \tau)], \quad S_{xy}(\omega) = \mathcal{F}\{R_{xy}(\tau)\} = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j\omega\tau} d\tau.$$

Also adopt the two-sided Wiener–Khinchin convention

$$R_{xx}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{j\omega\tau} d\omega,$$

under which unit-power white noise satisfies $S_{xx}(\omega) = 1 \iff R_{xx}(\tau) = \delta(\tau)$.

By linearity of expectation,

$$R_{xy}(\tau) = \mathbb{E}[x(t)v(t + \tau)] + \mathbb{E}[x(t)v^2(t + \tau)].$$

Using the convolution form of $v(t + \tau)$,

$$\mathbb{E}[x(t)v(t + \tau)] = \int_{-\infty}^{\infty} h(\lambda) \mathbb{E}[x(t)x(t + \tau - \lambda)] d\lambda = \int_{-\infty}^{\infty} h(\lambda) R_{xx}(\tau - \lambda) d\lambda.$$

Since $R_{xx}(\tau) = \delta(\tau)$,

$$\mathbb{E}[x(t)v(t + \tau)] = \int_{-\infty}^{\infty} h(\lambda) \delta(\tau - \lambda) d\lambda = h(\tau) = e^{-\tau}u(\tau).$$

Proof. Since $x(t)$ is zero-mean Gaussian white noise and $v(t + \tau)$ is a linear functional of $x(\cdot)$, the random vector $(x(t), v(t + \tau))$ is jointly Gaussian and zero-mean. By Isserlis' (Wick's) theorem, all odd-order central moments of a zero-mean jointly Gaussian vector vanish. In particular,

$$\mathbb{E}[x(t)v^2(t + \tau)] = \mathbb{E}[XV^2] = \mathbb{E}[X]\mathbb{E}[V^2] + 2\mathbb{E}[XV]\mathbb{E}[V] = 0,$$

since $\mathbb{E}[X] = \mathbb{E}[V] = 0$.

Therefore,

$$R_{xy}(\tau) = h(\tau) = e^{-\tau}u(\tau).$$

□

By definition, $S_{xy}(\omega) = \mathcal{F}\{R_{xy}(\tau)\}$. Since $R_{xy}(\tau) = h(\tau)$ and $h(t) = e^{-t}u(t)$, its Fourier transform is

$$H(j\omega) = \int_0^{\infty} e^{-t} e^{-j\omega t} dt = \frac{1}{1 + j\omega}.$$

For unit-power white noise, $S_{xx}(\omega) = 1$, hence

$$S_{xy}(\omega) = H(j\omega) = \frac{1}{1 + j\omega}.$$

Note that the input-output cross-correlation of this cascade system when the input is Gaussian White Noise with unity power level (interpreted as two-sided PSD $S_{xx}(\omega) = 1$, equivalently $R_{xx}(\tau) = \delta(\tau)$, in the generalized distribution sense).

Proof. By definition, $S_{xy}(\omega) = \mathcal{F}\{R_{xy}(\tau)\}$. From Problem 2, $R_{xy}(\tau) = h(\tau)$.

Thus, although the cascade system is nonlinear due to the static term $v^2(t)$, the input-output cross-correlation coincides with that of the LTI block alone for zero-mean Gaussian input. The Fourier transform of $h(t) = e^{-t}u(t)$ is $H(j\omega) = \int_0^{\infty} e^{-t} e^{-j\omega t} dt = (1 + j\omega)^{-1}$. $\square$

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